



“What Should I Do First?” The Effect of Manipulated Goal Conflict on Affect, Motivation, and Helping Behavior in Chronic Pain Couples

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Abstract: Although support provision by a partner is an important resource for individuals with chronic pain (ICPs), it poses a challenge for partners because it competes with other important personal goals of partners. The current study examined the impact of experimentally induced goal conflict in partners on their motives for helping, quality of provided help, and on partners' and ICPs' affect.

Sixty-eight couples, with at least one person having chronic pain, performed 2 series of household activities, with partners either asked to be simply available for help (ie, control condition) or to additionally work on a puzzle task (ie, goal conflict condition). Couples reported on interpersonal (eg, helping motives) and intrapersonal (eg, affect) outcomes. In addition, quality of partners' helping behavior and ICPs' pain behavior were videotaped and coded afterward.

In the goal conflict condition, ICPs were less satisfied with the received help and they experienced more pain. Also, the quality of the provided help was lower and partners experienced less positive and more negative affect. Addressing partners' goal conflict in clinical practice may help to avoid its negative impact on both ICPs and partners.

Perspective: This article provides a compelling argument to include partners in chronic pain treatment by demonstrating the detrimental effects of partners' experienced conflicts in goals upon the quality of help they provide, partners' affective functioning, and ICPs' pain-related outcomes.

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Introduction

Social support, especially from romantic partners, is an important resource for individuals with chronic pain (ICPs).⁴ Providing sufficient and high-quality help may, however, be a challenge for partners.^{17,36} Research has shown that some helping behaviors (eg, solicitous behaviors) may have unfavorable effects on patient

outcomes^{4,8,26,36,37} and that partners may appraise their helping role as stressful, which depletes their ability to provide effective support.^{3,23,56}

Drawing upon Self-Determination Theory (SDT),¹¹ the motives for providing help may explain when help can have an adaptive or a maladaptive effect on both ICP and partner. When help is provided for controlled or pressured motives (eg, avoidance of guilt/criticism), it has been found to predict poorer individual outcomes in both ICPs (eg, less satisfaction with help) and partners (eg, more helping exhaustion) as well as worse relational outcomes (eg, more relational conflict).^{28,30-32} Instead, when partners provide help for autonomous or volitional reasons (eg, commitment, enjoyment), they are more receptive to the preferences and needs of the ICP,²⁰ enabling them to provide more attuned and qualitatively better help.

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At present, there is a paucity of studies that identify factors that put partners at risk for developing controlled helping motivation. One such factor may be the conflict partners experience between their goal to help the ICP and the other goals they pursue on a daily basis (eg, performing well at work). Goal theories posit that each human being pursues multiple goals simultaneously and has to shift priorities between goals in order to function well.^{1,38} According to SDT,¹¹ the provision of daily support to the ICP may create pressure as the partner may feel conflicted about dividing his/her limited time and energy across different goals. If helping one's partner interferes with other tasks, it can be expected that the support provision might also be perceived more as a 'should', that is, a daunting duty that one feels compelled to do (ie, controlled motivation) rather than a task in which someone is engaged willingly (ie, autonomous motivation).²⁹ Such perceived pressure may yield an immediate affective cost (eg, enhanced negative affect). Moreover, goal conflict may engender lower quality of provided help, ie, help that is less attuned to the ICPs' basic psychological needs (ie, autonomy, competence, and relatedness), ultimately leading the ICP to be less satisfied with the received help.⁵¹ Goal conflict has been related to worse intrapersonal outcomes, such as lower subjective well-being^{16,39} and less successful goal attainment.⁶ Although research on interpersonal correlates of goal conflict is still in its infancy,⁵¹ there is preliminary, cross-sectional evidence for goal conflict to be related to lower relationship quality.¹⁶

In the present experimental-observational, multi-informant study, the causal effects of goal conflict on partners' helping behaviors and ICP and partner outcomes were examined. Chronic pain couples were invited to the lab, where ICPs were asked to execute different household tasks in the presence of their partner. While partners in the control group were asked to simply be available for the ICP to provide help, partners in the goal conflict condition were instructed to additionally perform a cognitive task. We expected that induced goal conflict may come with an interpersonal and intrapersonal cost. Interpersonally, goal conflict may undermine partners' volitional helping motivation, the quality of the provided help and satisfaction with the received help (even when controlling for relationship quality). Intrapersonally, goal conflict may cause reduced positive and enhanced negative affect in partners and ICPs, with ICPs also reporting more pain intensity and evidencing more pain behaviors. Outcomes were assessed through self-reports provided by partners and ICPs, and the displayed helping behavior by partners and pain behavior by ICPs were also recorded and coded.

Material and Methods

Participants

Participants were couples, recruited through the Flemish League for Fibromyalgia Patients (FLFP), a

patient organization for individuals with fibromyalgia. Members of the FLFP (N = 1391) received an invitation letter to participate in studies about chronic pain and quality of life in our lab in February 2015. About 35% (N = 481) agreed to be contacted by phone. Three hundred eighty-two of them indicated that they were currently in a romantic relationship. Inclusion criteria for participation of ICPs in the present study were (1) having chronic pain for at least 3 months, (2) physically living together with a partner for at least one year and (3) being sufficiently proficient in Dutch. From the couples that were contacted by phone (N = 344), 245 of the couples were reached, of which 30 did not meet the inclusion criteria, 130 did not want to participate mainly because of the distance to our lab, or because partners had no time. In total, 85 couples agreed to participate in the study. Seventy-six of them filled in a baseline questionnaire 2 weeks before the experiment and, finally, 72 couples came to the psychology building of Ghent University to take part in the experiment. Four couples were excluded from the analyses: data from 2 couples were used as pilot data, one couple terminated participation in the middle of the experiment and one couple turned out not to be living together (for an overview of recruitment details, see Fig 1). The final sample in this study included 68 couples.

The majority of ICPs were female (N = 62; 91.2%). The mean age of ICPs and their partner was, respectively, 49.68 (SD = 9.63) and 50.87 (SD = 9.97) years. All couples were Caucasian and 4 out of 10 (42.6% of ICPs; 38.2% of partners) reported an education beyond the age of 18. Eighty percent (N = 53) of the couples were married or legally cohabiting, with the mean relationship duration being 20.20 years (SD = 12.47). The majority of the partners were employed (N = 50; 73.5%), while only 27.9% of ICPs (N = 19) was employed. All ICPs reported at least 3 different pain locations (M = 5.51, SD = 1.23; range 3–7), with pain in the back, neck, and lower extremities being reported most frequently (all > 97%). Mean pain duration was 13.03 years (SD = 11.42). On a scale from 0 to 10, ICPs reported a mean pain intensity of 7.29 (SD = 1.00) and a mean disability of 6.45 (SD = 1.79). ICPs' mean pain duration was 13.55 years (SD = 10.03). Thirty-seven partners (ie, 54.4%) also reported pain complaints during the past 3 months (which is similar to other studies with chronic pain couples; eg, ²²). Paired samples *t*-tests showed that pain intensity (M = 3.74, SD = 1.94) and disability (M = 2.60, SD = 2.06) were significantly lower in partners compared with the ICPs (*ps* < .01).

Experimental Design

Couples attended a single lab session during which they were asked to perform 2 series of household activities together.⁴² Using a within-person design, goal conflict in partners was manipulated in a random and counterbalanced manner. In the control condition, partners were asked to be available for providing help to the ICP during the household activities, whereas in the goal conflict condition partners were asked to simultaneously perform a puzzle task. Because of the within-

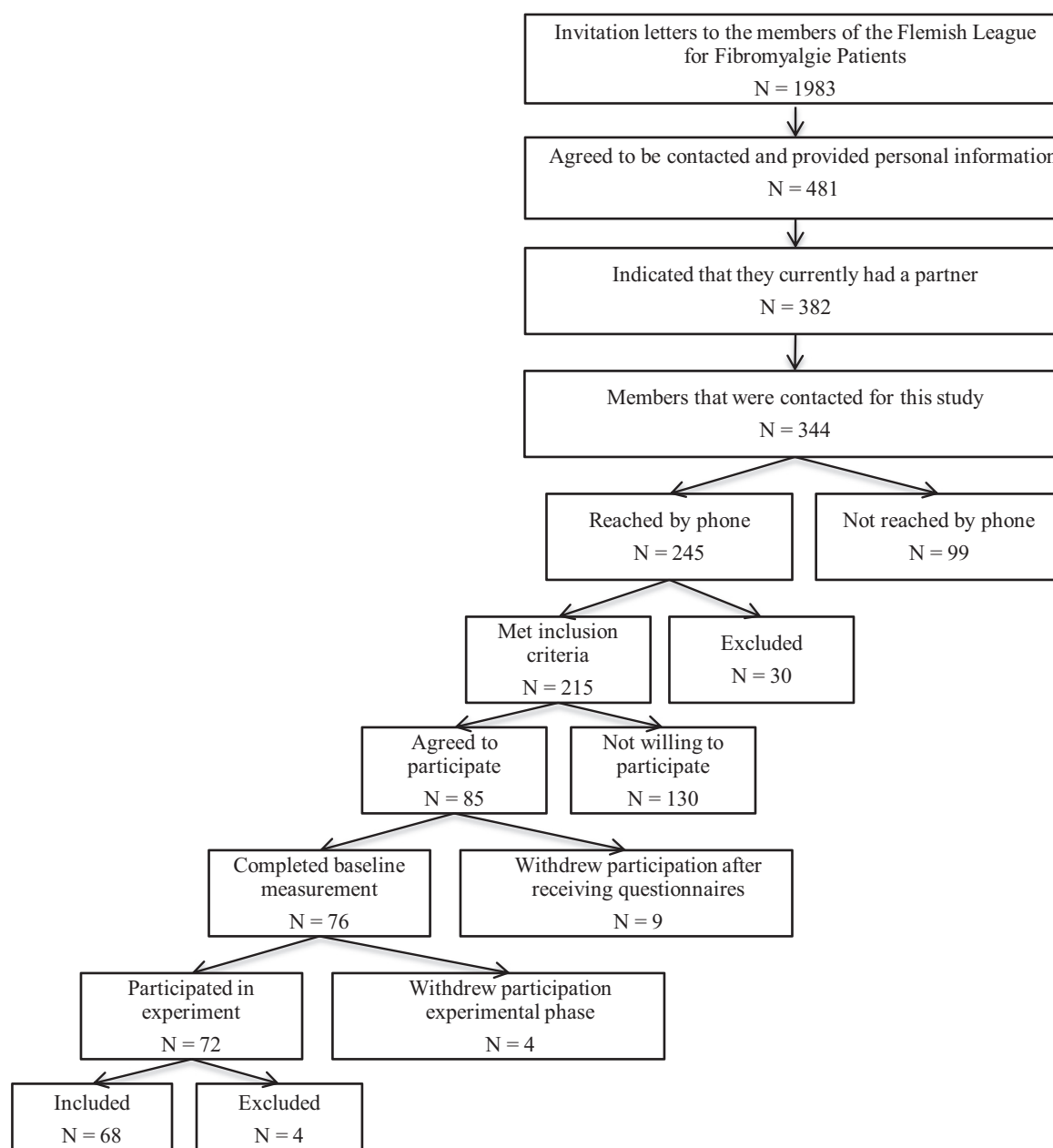


Figure 1. Flowchart of how final sample size was obtained.

subject design, each couple participated in both conditions, as we used a repeated measurement design. All ICPs are placed in both the control condition and the goal-conflict condition (yet in a counterbalanced way). The series of household activities differed between the 2 conditions in order to avoid familiarity with the household tasks.

Upon arrival at the lab, couples were told that the study aimed at investigating how couples handle household activities. Before the experiment started partners and ICPs were separated to answer a series of sociodemographic questions individually. Additionally, in the goal conflict condition, partners were instructed how to solve the puzzle task. After each series of household activities, couples again separately filled out another set of questionnaires with task-specific measures (eg, affect,

helping motivation, need supportive helping behaviors).

Measures

Self-Report Measures Completed Before the Experimental Session

Relationship quality. To control for the effects of general relationship quality, relationship quality was assessed before the experimental session using the short version of the Dyadic Adjustment Scale (DAS-7).^{21,44} The DAS-7 consists of 3 items for dyadic consensus (ie, the extent of agreement between partners), 3 items for dyadic cohesion (ie, shared interests and activities) and 1 item for dyadic satisfaction.

Higher sum scores represent higher levels of relationship quality. In this study, Cronbach's alphas were .86 for ICPs and .83 for partners.

Self-Report Measures During the Experiment

Socio-demographic variables. Both partner and ICPs reported on gender, age, nationality, marital status, relationship duration, having children, education, and professional status.

Manipulation check. After each series of household tasks, partners were asked to what extent the following statements were true (on a scale ranging from 0 "not at all true" to 6 "totally true"): "Time prevented me to fully carry out my tasks", "I felt pressured while performing my tasks", "I felt like having to do too many things at once" and "It was difficult to carry out the assigned tasks as good as I wanted to". We calculated a mean score of these 4 items to have a measure for our manipulation check. Cronbach's alphas were .82 (goal conflict condition) and .79 (control condition).

Affect. Both partners and ICPs reported on how they felt during the performance of the household activities by rating 6 items tapping into positive affect (eg, proud, happy, relaxed) and 6 items tapping into negative affect (eg, sad, nervous, scared),¹⁵ that have been used in previous studies in the context of chronic pain.³² Items on a 7-point scale ranged from 0 (totally disagree) to 6 (totally agree). Scores were computed by averaging items measuring positive and negative affect, respectively (for Cronbach's alphas, see Table 1).

Pain intensity. After each series of activities, ICPs reported on their experienced pain intensity (ie "How intense was your pain during the performance of these

tasks?") on a 7-point scale from "0" (no pain) to "6" (worst imaginable pain).

Helping motivation. To measure partners' helping motivation during the tasks, we used the Motivation to Help Scale⁵² that was successfully adapted in a previous study for use with chronic pain couples.³¹ Eight items, that is, 4 items assessing controlled motives (eg, "because I would feel guilty if I didn't help" or "because my partner demanded it from me") and 4 items assessing autonomous motivation (eg, "because I enjoy helping my partner" or "because I think it is important to help my partner"), were selected based on previous research with chronic pain couples.³² Partners rated these 8 reasons for supporting their partner in pain during the performance of the household activities on a 7-point scale ranging from "0" (not at all true) to "6" (totally true). ICPs received the same list of reasons and reported on the *perceived* helping motivation of their partner. For perceived helping motivation Cronbach's alphas were rather low, but this is not surprising because rating an introspective process as an observer is not an easy task. For all Cronbach's alphas, see Table 1.

Quality of helping behavior. In line with SDT,⁴³ we operationalized quality of helping behavior as need-supportive helping behavior, ie, helping that addresses the basic psychological needs of the ICP. To measure need supportive helping behavior (as self-reported by the partner and perceived by the ICP), we developed a 6-item scale based on prior work in health care settings.⁵³ In line with recent research,⁴⁰ 2 autonomy-supportive (AS; eg, "asked if my partner needed help"), 2 competence-supportive (CS; eg, "made useful suggestions about how to handle these tasks") and 2 relatedness-supportive (RS; eg, "paid attention to my partner

Table 1. Correlations Between Self-Report and Observational Measures in the Goal Conflict Condition (Below Diagonal) and Control Condition (Above Diagonal)

VARIABLES	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 AM ^{ppt}	—	.05	.62***	-.18	.56***	.23	.10	.02	.08	.24	.27*	.14	-.06	.28*
2 CM ^{ppt}	.15	—	-.03	.48**	.19	-.11	.33**	.15	.05	.08	.17	.08	-.06	-.07
3 PA ^{ppt}	.61***	-.07	—	-.22	.28*	.24	.22	.19	-.01	.09	.26*	.07	-.25*	.04
4 NA ^{ppt}	-.03	.28*	-.25*	—	-.00	-.21	-.03	-.05	-.01	.03	-.05	-.00	.12	.05
5 NSUPP ^{ppt}	.54***	.26*	.11	-.02	—	.04	.12	.09	.05	.18	.13	.04	.15	.33**
6 AM ^{icp}	.16	.06	-.05	-.10	-.04	—	.24	.29*	-.09	.35**	.24	.26*	-.05	-.03
7 CM ^{icp}	.07	.33**	.12	-.12	.08	.18	—	.13	.10	.14	.16	.07	-.4	-.18
8 PA ^{icp}	-.05	.16	.04	.00	-.09	.41**	.30*	—	-.45***	.09	-.02	-.04	-.19	.03
9 NA ^{icp}	.26*	.23*	.20	-.09	.19	.10	.21	-.14	—	.04	.03	.42***	.12	-.22
10 NSUPP ^{icp}	.39**	.07	.05	-.03	.36**	.48***	-.08	.11	.09	—	.33**	.33**	.16	.36**
11 SRH ^{icp}	.51***	.15	.23	.02	.34**	.25	-.04	-.07	.06	.67**	—	.23	.17	-.01
12 Pain ^{icp}	.16	.03	.08	.08	.10	.02	-.16	-.23	.45***	.14	.22	—	.26*	-.02
13 Pain Beh ^{obs}	.02	.09	-.09	.13	.01	-.20	.00	-.37**	.34**	-.12	-.18	.36**	—	.04
14 NSUPP ^{obs}	.46***	.32*	.17	.00	.57***	.18	.01	-.02	.12	.50***	.35**	.00	-.05	—
α Goal conflict condition	.91	.70	.86	.75	.60	.83	.58	.78	.75	.72	/	/	/	/
α Control condition	.82	.57	.86	.69	.59	.81	.52	.79	.80	.58	/	/	/	/

Abbreviations: ppt, partner; ICP, individual with chronic pain; obs, observed; AM, autonomous helping motivation; CM, controlled helping motivation; PA, positive affect; NA, negative affect; NSUPP, need supportive helping behavior; SRH, satisfaction with received help; Beh, behavior.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

and was involved") items were included which were rated on a 7-point Likert scale, ranging from "0" (not at all) to "6" (extremely). Items were slightly adapted to tap into ICPs' perceived need support. A need-supportive helping behavior scale was created by averaging all 6 items and was found to be reliable with acceptable Cronbach's alphas (see Table 1) among both partners and ICPs.

Satisfaction with received help. After each series of household tasks, ICPs also reported on their satisfaction with the received help (ie, "to what extent are you satisfied with the received help from your partner?") on a 7-point scale from "0" (not at all) to "6" (extremely satisfied).

Observational Assessment

Observational data were collected by videotaping couples while engaging in the household activities. All interactions were broken down into 10-second intervals, with a mean of 35.69 (SD=4.91) rated intervals in the goal conflict condition and 32.58 (SD=6.33) rated intervals in the control condition. Within each interval, partners' quality of provided help (ie, level of need support) was coded, together with ICPs' pain behaviors. A coding manual was available and an intensive training with 10 videos were provided to coders. Both coders watched these videos together, rated videos individually, followed by a discussion of the differences in coding scores. After this training phase, 20% of the videos were coded by 2 raters, while the remaining 80% of the videos were coded by one rater only (ie, each 40%). Inter-rater reliability was calculated according to the formula provided by Ekman and Friesen¹⁴ that assesses the proportion of agreement¹ by 2 coders relative to the total number of actions coded by each coder. It was not possible for the raters to be blind to the experimental condition as this condition could be recognized by the fact that partners were performing the puzzle task.

Quality of helping behavior. A coding scheme was developed to assess need-supportive helping behavior and was similar to the items used in our self-report scale for partners and ICPs. In other research contexts, ie teaching^{46,19} and parenting,⁵⁵ observed need support has been reliably coded. In a first phase, a dozen videos were watched by author S.K. and another independent researcher. Based upon the self-report items and expert opinion, an initial set of items for use in the coding scheme was formulated. In a second phase, 2 raters (ie, author S.K. and another research assistant) coded the first five videotapes. Based on their experiences, some problems were identified (eg, low frequency of some of the behaviors) and the coding scheme was adapted. In the final coding scheme, a distinction was made between the presence ("1") or absence ("0") of Autonomy-supportive (AS), Competence-supportive (CS), and

Relatedness-supportive (RS) helping behaviors. AS was present when the partner asked if the ICP needed help, or when s/he provided a choice how to handle the tasks for example by letting the ICP take the lead. CS was present if the partner gave positive feedback or useful suggestions. RS was present when the partner was listening, attentive and involved. Involvement could comprise actively helping the ICP but also asking questions about the task or physical contact between the partner and the ICP. For each 10-second interval, the coder scored the presence ("1") or absence ("0") of each of these 6 behaviors. All subscales showed good inter-rater reliability¹⁴ ranging from .76 to .93. To obtain a more general measure for the quality of help we calculated an overall score of Need Support (NS) by summing up AS, CS and RS; excellent inter-rater reliability was achieved for this overall score (.95).

ICP pain behavior was coded based on a scheme used in previous studies with ICPs.¹³ Each 10-second time interval was coded for the presence ("1" slightly present; "2" distinctly present") or absence ("0") of different types of pain expression: facial pain expression (eg, narrowing eyes, raising cheeks), paraverbal (eg, sighs or moans) and verbal pain behavior (using pain words, such as 'ouch'), active pain behavior (eg, guarding, holding, or rubbing) and passive pain behavior (ie, interrupt or avoid performing task). The different types of pain behavior were not rated separately, the inter-rater reliability for ICP pain behavior was acceptable (.69).

Procedure

Two weeks before the experiment, participants who were willing to participate in the experiment were sent an e-mail with the link to an online questionnaire (including the DAS-7) and informed consent. After completing the questionnaire, couples were invited to the faculty for the experimental phase of the study, consisting of 2 different experimental studies. The first experiment focused upon partners' goal conflict (reported in this paper) and the second experiment, which took place subsequently, will be reported elsewhere. In this second experiment, a Cyberball paradigm was used to examine the buffering effect of having a partner present on feelings of social exclusion by the patient. There is no overlap between the data of these 2 experiments. Before the experiment, couples were asked permission to be videotaped. At the end of these 2 experiments, couples were debriefed about the goals of the study and the reason why an extra puzzle task was given to partners and were asked, a second time, permission to use the observational data for educational or scientific purposes. As a sign of appreciation, couples received a 30€ fee after participation in both experiments. This study was approved by the ethical committee of the Faculty of Psychology and Educational Sciences of Ghent University.

Household Activities

The 2 different series of household activities (ie, series A and B), each consisting of 4 activities, were

counterbalanced across the 2 experimental conditions. All ICPs performed both series A and B, but the order in which they performed these series was randomly assigned. Both series included changing linens on a bed, picking up small objects and carrying a heavy bag, but these activities were performed in a different and set order. Furthermore, in series A participants were additionally asked to cover a table with a tablecloth, while in series B participants were asked to hang coats on a coat stick. These tasks, slightly adapted from earlier work of Romano and colleagues,^{41,42} were chosen because they elicit pain behaviors in ICPs and allow the ICP and partner to interact together. Couples received an instruction sheet detailing the order in which the tasks had to be performed. They were allotted 6 minutes to complete the task, which prevented couples from dawdling. ICPs were asked to take the lead during the activities, but they were informed that they could ask their partner for help whenever they wanted to.

Manipulation of Goal Conflict

Separate instructions were given to partners prior to the goal conflict or control condition. In the control condition partners were asked to remain available for help during the household activities, in case help was requested by the ICP. In the goal conflict condition, partners were told that they received an extra task. This task consisted of making as many puzzles as possible on a tablet, while they were instructed to stay available to provide help to the ICP during the execution of household activities. As a cover story, partners were told that we aimed to ascertain their ability to multitask, a skill which is often needed in daily life. To further induce goal conflict, partners were informed that individuals who were proficient in multitasking were often seen as more intelligent, competent, and creative and they were informed that they would receive their own multitasking scores at the end of the experiment. In both conditions partners were told that they were free to choose to help their partner or not (see Appendix for full instructions). During the performance of the household tasks, the research assistant waited outside the experiment room. After each series of household activities, partners and ICPs completed the same set of questionnaires, which included a manipulation check.

Data Analytic Strategy

First, for all observational data, sum scores were divided by the number of coded time intervals, to control for the variation in length of the videos. Subsequently, Pearson's bivariate correlations were conducted to examine the relationships between the included self-report variables and the observational data in both conditions (see Table 1).

Second, repeated measures ANOVA were conducted to examine the effect of goal conflict on different partner and ICP outcomes. Outcome variables were divided in interpersonal outcomes (ie, helping motivation, quality of help in terms of need supportive helping behavior,

satisfaction with received help) and intrapersonal outcomes (ie, ICP and partner affect, ICP pain intensity, ICP pain behavior). Analyses were repeated controlling for the order in which conditions were offered (control condition – goal conflict condition or vice versa), indicating a similar pattern of results, which we will not report in the result section. As general relationship quality (DAS-7) may influence interpersonal outcomes (ie, helping behavior and motivation to help), relationship quality was added as a covariate.

Results

Correlations of Helping Motivation With Outcomes in Both Conditions

Correlations between self-report and observational measures are presented in Table 1, for the goal conflict condition (below the diagonal) and the control condition (above the diagonal). Volitional or autonomous helping motivation was found to relate with several interpersonal outcomes. Specifically, partner-reported volitional helping motivation related positively to ICPs' satisfaction with received help and need-supportive helping behavior, as reported by partners, observed by coders or reported by ICPs. These associations emerged in both conditions, except for the nonsignificant relation with ICP-reported helping behavior in the control condition. The volitional helping motivation as perceived by the patient was only significantly related to ICP-reported need support, yet in both conditions. In contrast, controlled helping motivation (as reported by either the partner or the ICP) was unrelated to any interpersonal outcome, with 2 exceptions. Controlled helping motivation reported by the partner related positively to partner-reported and observed need-supportive helping behavior, yet only in the goal conflict condition.

Impact of Goal Conflict on Interpersonal Outcomes

Results of the repeated measures ANOVA, can be found in Table 2. Preliminary analyses showed that partners' responses on the manipulation check items were significantly different in the goal conflict relative to the control condition ($F(1,67) = 43.33$, $P < .001$; $\eta^2_{\text{partial}} = .39$). Partners found it more difficult to handle both tasks and experienced more pressure when being in the goal conflict, relative to the control condition, indicating that the manipulation was successful. For all interpersonal outcomes, relationship quality (as measured by the DAS-7) was added as a covariate in the analyses.

For all helping motivation measures, no significant effect of goal conflict was found. Partners' self-reported autonomous and controlled helping motivation, as well as partners' autonomous and controlled helping motivation as perceived by the ICP did not differ between conditions ($p > .05$).

Table 2. Repeated Measures ANOVA Indicating the Effects of Goal Conflict Upon Interpersonal and Intrapersonal Outcomes

DEPENDENT VARIABLES	INFORMANT	WITHIN-SUBJECT EFFECTS CONDITION		DESCRIPTIVE M (SD)	
		F	PARTIAL η^2 (ES)	GOAL CONFLICT	CONTROL
Manipulation check	Partner	43.33***	.39	1.85 (1.45)	.72 (1.01)
Interpersonal outcomes					
Autonomous helping motivation	Partner [#]	.15	.002	3.78 (1.65)	3.94 (1.41)
	ICP ^o	2.93	.001	3.54 (1.54)	3.91 (1.45)
Controlled helping motivation	Partner [#]	.07	.001	1.36 (1.25)	1.48 (1.11)
	ICP ^o	.24	.004	1.31 (1.14)	1.42 (1.10)
Quality of help (Need support)	Partner [#]	.05	.001	2.79 (.93)	3.08 (1.00)
	ICP ^o	10.10**	.14	2.65 (1.23)	3.39 (1.01)
	Observer ^o	24.22***	.28	.50 (.26)	1.03 (.18)
Satisfaction received help	ICP ^o	4.89*	.07	4.72 (1.44)	5.16 (.88)
Intrapersonal outcomes					
Positive affect	Partner	4.97*	.07	3.13 (1.33)	3.38 (1.35)
	ICP	3.70 [†]	.05	2.77 (1.15)	2.97 (1.21)
Negative affect	Partner	8.84**	.12	.50 (.69)	.32 (.51)
	ICP	.10	.002	1.20 (1.02)	1.17 (1.25)
Pain intensity	ICP	4.35*	.07	3.59 (1.23)	3.34 (1.13)
Pain behavior	Observer	4.40*	.06	.40 (.28)	.35 (.25)

Abbreviation: ICP, Individual with Chronic Pain.

[#]With partner-reported relationship quality as covariate. ^o With ICP-reported relationship quality as covariate.

[†] $P < .10$.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

The effects for need-supportive helping behavior (ie, indicating quality of help) were consistent across ICPs $F(1,64) = 10.10$, $P = .002$; $\eta^2_{\text{partial}} = .14$) and observers ($F(1,64) = 24.22$, $P < .0005$; $\eta^2_{\text{partial}} = .28$), respectively, experiencing, and observing less need-supportive behaviors in the goal conflict, relative to the control condition. The effects for need-supportive helping behavior reported by partners were not significant ($F(1,65) = 2.89$, ns). Finally, a main effect of goal conflict was found upon ICPs' satisfaction with received help ($F(1,61) = 4.89$, $P = .03$; $\eta^2_{\text{partial}} = .07$), with ICPs in the goal conflict condition being less satisfied with the help they received from their partner.

Furthermore, partner-reported relationship quality predicted their volitional motivation to provide help ($F(1,64) = 9.44$, $P = .003$; $\eta^2_{\text{partial}} = .13$), while being unrelated to their controlled motivation to provide help ($F(1,64) = .16$, ns). Similarly, patient-reported relationship quality predicted their perception of autonomous helping motivation by their partner ($F(1,54) = 9.29$, $P = .004$; $\eta^2_{\text{partial}} = .15$), while being unrelated to perceived controlled helping motivation by their partner ($F(1,54) = 1.49$, ns). Finally, while partner-reported relationship quality did not significantly predict partner-perceived need-supportive behavior ($F(1,66) = 1.51$, ns), patient-reported relationship quality significantly predicted patient-perceived need-supportive helping behavior ($F(1,64) = 20.80$, $P < .0005$; $\eta^2_{\text{partial}} = .25$) and satisfaction with the received help ($F(1,61) = 19.02$, $P < .0005$; $\eta^2_{\text{partial}} = .24$), but did not significantly predict observed need-supportive behavior ($F(1,64) = 1.52$, ns).

Impact of Goal Conflict on Intrapersonal Outcomes

When placed in the goal conflict condition, partners experienced lower positive affect ($F(1,65) = 4.97$, $P = .03$; $\eta^2_{\text{partial}} = .07$) and higher negative affect ($F(1,65) = 8.84$, $P = .004$; $\eta^2_{\text{partial}} = .12$), while a marginally significant effect of goal conflict on ICPs' positive affect was found ($F(1,65) = 3.70$, $P = .06$; $\eta^2_{\text{partial}} = .05$). Results further indicated a significant main effect of condition on ICPs' self-reported pain intensity during the household tasks ($F(1,63) = 4.35$, $P = .04$; $\eta^2_{\text{partial}} = .07$) and on ICPs' observed pain behavior ($F(1,66) = 4.40$, $P = .04$; $\eta^2_{\text{partial}} = .06$). ICPs experienced higher pain levels and were observed to display more pain behavior in the goal conflict condition compared with the control condition.

Discussion

ICPs benefit from daily support by their partner.⁴ Previous research has indicated that a volitional motivation to provide help may yield the most beneficial effects,^{27,28,30-32} and that this may be threatened when partners experience conflicts between the goal to help the ICP and other valuable goals they pursue in daily life.²⁹ Providing help requires one's limited time and energy. The aim of the present study was to mimic this daily experience within an experimental setting as to provide causal evidence of the intrapersonal and interpersonal costs associated with induced goal conflict. In

detailing these costs, we probed both ICPs' and partners' perspective and coded partners' helping behavior and ICPs' pain behavior.

Interpersonal Functioning: Quality of and Satisfaction With Help

First, we examined the impact of goal conflict on the quality of partners' helping behavior. Our first indicator of quality of help was partners' helping motivation. According to Self-Determination Theory (SDT),¹¹ when partners are autonomously motivated to provide help, they experience it as more enjoyable, personally valuable and meaningful, while, in the case of controlled helping motivation, they provide help to avoid the recipients' criticism or feelings of guilt or to garner external approval. Abundant research in other contexts has found autonomous, relative to controlled, motivation to yield manifold benefits, including behavior persistence, enhanced well-being, and better relationship functioning.^{43,48} In this study, goal conflict did not affect the helping motivation as reported by partners and ICPs. This is not in line with a previous diary study in the context of chronic pain,²⁹ where daily goal conflict was related to lower autonomous, relative to controlled, helping motives from day-to-day. General levels of relationship quality were related to more autonomous helping motivation during the household tasks, indicating that helping motivation may be more strongly influenced by general perceptions of relationship quality than by situational variations in goal conflict. Maybe the goal conflict manipulation was less hurtful in the case both partners get along very well. Future research could try using an implicit measure of motivation, which may be potentially more context-sensitive and less susceptible to social desirability response tendencies. Considerable validation work will, however, be necessary.⁴⁷

The limited effects of goal conflict on the motivational outcomes stand in contrast to how partners behaved in the presence versus absence of goal conflict. The help provided by the partner was perceived (ie, ICP) and observed to be less need-supportive during goal conflict. Taken together, goal conflict seems to affect to a lesser extent the motivational basis for providing help but instead yields a more direct impact on the type of help being provided. Interestingly, helping motivation was also related to the quality of help provision. More partner-reported volitional helping motivation was related to more partner-perceived and observed need support, more ICP-reported need support (only during goal conflict) and more satisfaction with received help.

According to SDT, partners' provision of need support, that is, help which is attuned to ICPs' needs for autonomy, competence and relatedness, is said to be critical. The satisfaction of these needs is considered essential for individuals to grow and reach their full potential.¹¹ Depending on the degree to which these needs get satisfied or frustrated, one can reliably predict differences, both interpersonally as well as intrapersonally, in well-being, (mal)adjustment and even

psychopathology.⁴⁹ Partners can be more or less supportive of ICPs' needs. In this study, partners appeared more likely to ask the ICP whether or not they were in need for help (ie, autonomy support), provide positive feedback (ie, competence support) and be attentive to what the ICP was doing (ie, relatedness support) in the absence, relative to the presence, of goal conflict. The fact that partners provided less qualitative help can explain why ICPs were less satisfied with the provided help in the goal conflict condition. Noteworthy, ICP-perceived need support and satisfaction with the help they received were also predicted by ICPs' general perceptions of relationship quality, irrespective of goal conflict. This indicates that both general perceptions about the relationship as well as situational demands may determine how the help by partners is perceived by ICPs.

These findings are indirectly in line with a recently proposed affective-motivational account of interpersonal dynamics in the context of pain.⁵¹ This model states that individuals with high self-oriented goals (eg, focus on own goals when experiencing goal conflict), rather than other-oriented goals, when faced with another in pain, may display diminished feedback sensitivity that impedes the receptivity or attention to the needs of the person in pain, potentially contributing to rigid or inflexible caregiving behavior.⁵¹

Intrapersonal Functioning: Affect and Pain

Goal conflict not only comes with an interpersonal but also an intrapersonal cost. In the goal conflict condition, partners reported lower well-being, while ICPs reported more pain intensity and were observed to display more pain behavior. Previous correlational work also found goal conflict to be negatively associated with individuals' subjective well-being.^{16,39} This finding is also in line with work of Kasser,²⁵ who reported that the feeling that one has enough time (ie time affluence) relates positively to subjective well-being.²⁵ Sufficient time is a critical resource to engage in activities that promote personal growth and connection with others, activities that typically enhance someone's well-being.²⁴ A lack of time may also more easily trigger feelings of pressure that may interrupt one's ability to be available in the moment⁷ and to experience "flow",¹⁰ both of which facilitate happiness. The goal conflict condition in this study created (time) pressure, probably diminishing partners' feeling of autonomy. Notably, the goal conflict induction not only affected partners' well-being but also ICPs' pain behavior. Under circumstances of goal conflict, ICPs perceived physical tasks as more painful and ICPs were observed to display more pain behavior. Possibly, this is due to the lower quality of help (ie, less support of ICPs' needs) provided by partners in these circumstances. As noted by several research groups,^{9,18,45} pain behavior may communicate the need for support or intimacy to close others. Indeed, pain

behavior may elicit greater emotional and instrumental support by way of signaling greater pain intensity to observers. Participants in the goal conflict condition may have engaged in more pain behaviors to more clearly communicate with their potentially distracted partners that they were in pain and desired some form of support.^{12,13}

Limitations

Some limitations deserve attention. First, the sample consisted predominantly of women. As noted by Revenson et al⁵, gender may affect support provision in number of ways including household division of labor and gendered expectations for emotional support. Continued research is required to understand how gender relates to support provision in an SDT context. Second, all couples were recruited from a self-help organization and may not be representative of couples in primary or tertiary care centers. Only 85 couples of 215 that were reached and met the inclusion criteria, agreed to participate in the study, which may reflect a selection bias. Third, although we tried to enhance ecological validity by imitating a living room, the generalizability of these results to the home setting is unclear. Fourth, most effect sizes were small in size. Given the impact of general relationship quality this is not totally unexpected. Future research with a larger sample should aim to replicate the current pattern of findings before drawing firm conclusions. Finally, although previous research showed satisfactory internal consistencies,^{29,32} in this study they were a bit low, especially for controlled helping motivation.

Clinical Implications

Having conflicting goals is inevitable,³⁹ but given the potential detrimental effects upon the quality of helping, partners' affective functioning and ICPs' pain-related outcomes, this study provides a compelling argument to include partners into treatment. Partners may benefit from a clarification of their personal values and goals through Acceptance and Commitment Therapy.^{33,34} Through values clarification, partners may become aware of potential conflicts between valued

goals and learn ways to manage conflicts. Although caregiving could elicit feelings of sadness and helplessness, there is room for change to see this helping role as less interfering with other valued goals or even as an enrichment in life.⁵⁰

Furthermore, couples may be taught communication skills to discuss threats to the helping process, such as goal conflict experienced by partners, and to foster dyadic coping with pain.³⁵ For instance, ICPs may be taught to clearly communicate their needs to their partner. For ICPs and partners it is important to be aware of attempts of indirect support seeking, because this tends to be aversive for potential support providers,^{2,54} who may react with unsupportive or rejecting behaviors.

Conclusion

Having to combine another task with helping your partner in pain showed to be detrimental for the quality of provided help. Partners were less attuned to the needs of their partner in pain when a second task interfered with their helping role. Furthermore, an induction of conflicting goals in partners diminished partners' well-being and also induced ICPs' pain intensity and behavior during household tasks, potentially due to a lower quality of provided help. Addressing partners' goal conflict in clinical practice may be useful to avoid the negative impact on interpersonal and intrapersonal processes.

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Supplementary data

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